Deep Domain Fidelity and Low-Rank Tensor Ring Regularization for Thick Cloud Removal of Multitemporal Remote Sensing Images

Hao Peng, Ting-Zhu Huang, Xi-Le Zhao, Jie Lin, Wei-Hao Wu, *Student Member, IEEE*, and Li-Yuan Li

***Abstract*— Thick cloud contamination in remote sensing (RS) images significantly hinders their utility in subsequent applications. Traditional cloud removal methods predominantly focus on the design of the regularization term, while neglecting the design of the fidelity term. Recently, the proposal of gradient-domain fidelity has highlighted the significance of the fidelity term design, dedicated to maintaining textures in the gradient domain. However, the handcrafted gradient-domain fidelity still has limitations in capturing invariant and delicate features behind multitemporal RS (MTRS) images, leading to unsatisfactory detail preservation. To address the problem, we suggest a deep domain fidelity to capture deep features by leveraging a pretrained deep network, which matches the intrinsic deep feature between the original images and reference images, instead of matching the shallow features in gradient domain fidelity. Empowered with the deep domain fidelity, we propose a thick cloud removal model (called DFTR) for MTRS images, organically integrating the deep domain fidelity term with a low-rank (LR) regularization term (i.e., tensor ring (TR) decomposition), offering fine detail preservation. Extensive simulated and real experiments on MTRS images demonstrate that the proposed method outperforms the compared methods, including the origin domain-based and the gradient domain-based methods, in thick cloud removal, especially for detail preservation.**

***Index Terms*— Deep domain fidelity, multitemporal remote sensing (MTRS) images, tensor ring (TR) decomposition, thick cloud removal.**

NOMENCLATURE

Notations	Explanations
x, X, $\mathbf{x}$, $\mathbf{X}$, $\mathcal{X}$	Scalar, scalar, vector, matrix, tensor.
$\mathcal{X} \in \mathbb{R}^{H\times W\times B\times T}$	MTRS image, where H and W are the spatial resolutions, B and T are the number of the spectral bands and the time nodes, respectively.
$\mathcal{X}(i_1,\ldots,i_N)$	Element of tensor $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times\cdots\times I_N}$ in position $(i_1,\ldots,i_N)$.
$\mathcal{X}^t$	tth time node of $\mathcal{X} \in \mathbb{R}^{H\times W\times B\times T}$.
$\mathbf{X}_b^t$	bth spectral band and the tth time node of $\mathcal{X} \in \mathbb{R}^{H\times W\times B\times T}$.
$\mathcal{G}^{(n)} \in \mathbb{R}^{R_n\times I_n\times R_{n+1}}$	Third-order core tensor of Nth-order tensor $\mathcal{X} \in \mathbb{R}^{I_1\times I_2\times\cdots\times I_N}$ obtained by the circular multilinear product.
$\|\mathcal{X}\|_F$	Frobenius norm. $\|\mathcal{X}\|_F = (\sum_{i_1,\ldots,i_N} \|\mathcal{X}(i_1,\ldots,i_N)\|^2)^{1/2}$.

I. INTRODUCTION

AS REMOTE sensing (RS) technology rapidly evolves, multitemporal RS (MTRS) images, containing rich spatial, spectral, and temporal information, have been widely used in classification [1], [2], [3], target recognition [4], [5], [6], fusion [7], [8], [9], [10], and unmixing [11], [12], [13]. The presence of a thick cloud significantly disrupts optical satellite observation missions and subsequent applications. Therefore, it is essential to develop cloud removal methods for RS images to enhance RS interpretations and applications.

Various methods have been proposed for the removal of thick clouds from RS images. Considering that these methods utilize the rich spatial, spectral, and temporal characteristics inherent in MTRS images, they are divided into four primary types [14]: spatial-based, spectral-based, temporal-based, and hybrid-based methods.

The spatial-based methods primarily rely on the similarity of the structure and features of the cloud-free regions and the cloud-contaminated regions in the spatial dimension. According to the mechanism of the cloud-contaminated image reconstruction task, this problem can be converted into an image inpainting problem. Therefore, many inpainting methods can be applied to thick cloud removal [15], [16], which can efficiently recover the missing regions by calculating the weighted average of the neighborhood pixels. The downside is that it is difficult for them to get accurate reconstruction results for large missing regions. To further improve the reconstruction effect, exemplar-based methods [17], [18], variation-based methods [19], [20], and diffusion-based methods [21] are proposed. Nevertheless, these methods are limited by the use of insufficient reference information, and the reconstruction of large-scale cloud cover or areas with complex textures is still not satisfactory. The spectral-based methods offer an

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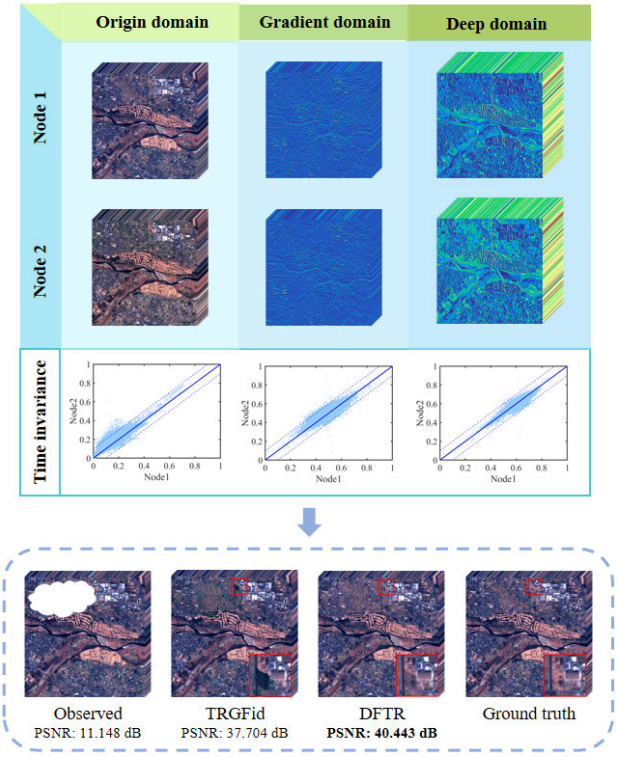


Fig. 1. Comparison of feature invariance with respect to different times in different transformed domains. The third row shows scatter diagrams of images at time nodes 1 and 2. The first column presents ground-truth images for the two nodes. The second column contains gradient feature maps, and the third column shows deep feature maps for the nodes. The last row displays the observed image, reconstructed images, and ground truth at time node 1.

effective perspective to reconstruct the RS images through the utilization of spectral information. For example, due to band 6 of the Aqua moderate resolution imaging spectroradiometer (MODIS) being contaminated by striping or noise, Wang et al. [22] proposed an approach to recover band 6 of Aqua MODIS from band 7 based on the relationship between bands 6 and 7 of Terra MODIS. Furthermore, Shen et al. [23] and Li et al. [24] took into account more information, such as scene information and all the other spectral bands, to further achieve better reconstruction results. Since thick clouds block all spectral bands of the signal resulting in no reference information between spectral bands, the spectral-based methods cannot achieve satisfactory results for thick cloud removal. The temporal-based methods [25] focus on analyzing the correlation between each time node in multitemporal images to reconstruct the cloudy region. Beck et al. [26] used a double logistic function for modeling the normalized vegetation index time series. Zhu et al. [27] presented a neighborhood-similar pixel interpolator approach that was developed to fill missing pixels. Subsequently, Zeng et al. [28] recovered the cloud-contaminated data based on the weighted linear regression from the MTRS images. The temporal-based methods provide a new perspective for thick cloud removal and have demonstrated effectiveness, but using temporal information alone remains inadequate.

The hybrid-based methods aim to leverage spatial–spectral–temporal complementary information to effectively capture the multidimensional characteristics of MTRS images.

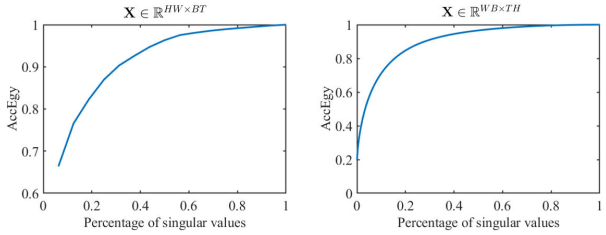


Fig. 2. Accumulation energy ratio (AccEgy : $\sum_j^k \sigma_j^2 / \sum_i \sigma_i^2$) of singular values of the circular unfolding of the original tensor $\mathcal{X} \in \mathbb{R}^{H \times W \times B \times T}$.

By integrating these diverse sources of information, these methods enhance the accuracy and quality of the reconstructed results. For instance, Chen et al. [29] combined the spatial and temporal information of Landsat images to establish a weighted regression model. Nevertheless, the effectiveness of cloud removal can be further improved by maximizing the use of existing information. Low-rank (LR) tensor decomposition methods are adept at exploiting the information across various dimensions in MTRS images and have received considerable attention in the reconstruction of MTRS images [30], [31], [32], [33], [34]. Ji et al. [35] analyzed the nonlocal correlations of RS images using the Tucker decomposition [36]. Ng et al. [37] presented the adaptive weighted tensor completion method for the reconstruction of RS images based on the weighted sum of the nuclear norms (SNNs) [38]. Based on the tensor singular value decomposition (t-SVD) [39], Wang et al. [40] exploited the low rankness of the RS image and leveraged the sparse structure of partial observations. Xu et al. [41] imposed the representation coefficient with a total variation (TV) regularizer for cloud removal. Chen et al. [42] proposed a tensor nuclear norm (TNN)-based [43] LR model with TV regularization. Zhang et al. [44] removed thick clouds in time-series images by combining deep prior and t-SVD. In addition, tensor network decomposition has performed well in MTRS images. Bengua et al. [45] introduced a matricization method via tensor train (TT) rank [46]. Yu et al. [47] proposed the LR model with smoothing of spatial and spectral dimensions based on TT decomposition and applied it to RS image recovery. Moreover, tensor ring (TR) decomposition [48] defined a well-balanced decomposition structure with a circulating tenet, which can integrate the spatial–spectral–temporal information. Taking the MTRS image from the *Tokyo* dataset as an example, Fig. 2 shows the accumulation energy ratio (AccEgy : $\sum_j^k \sigma_j^2 / \sum_i \sigma_i^2$) of singular values curves of the circular unfolding [5] of the original tensor. The first few singular values accommodate the most energy, indicating the LR TR structure of the MTRS images. He et al. [49] sought the smoothness of MTRS in the spatial dimension combined with the TR decomposition. These methods have achieved great results in pixel domain reconstruction. However, all of these LR models focus on the improving regularization term, but the fidelity term is neglected. Notably, Li et al. [50] introduced pioneering work in reconstructing MTRS images by characterizing textures based on gradient-domain fidelity with TR decomposition. While gradient-domain fidelity shows promise in preserving textures, it still has limitations in capturing essential features and

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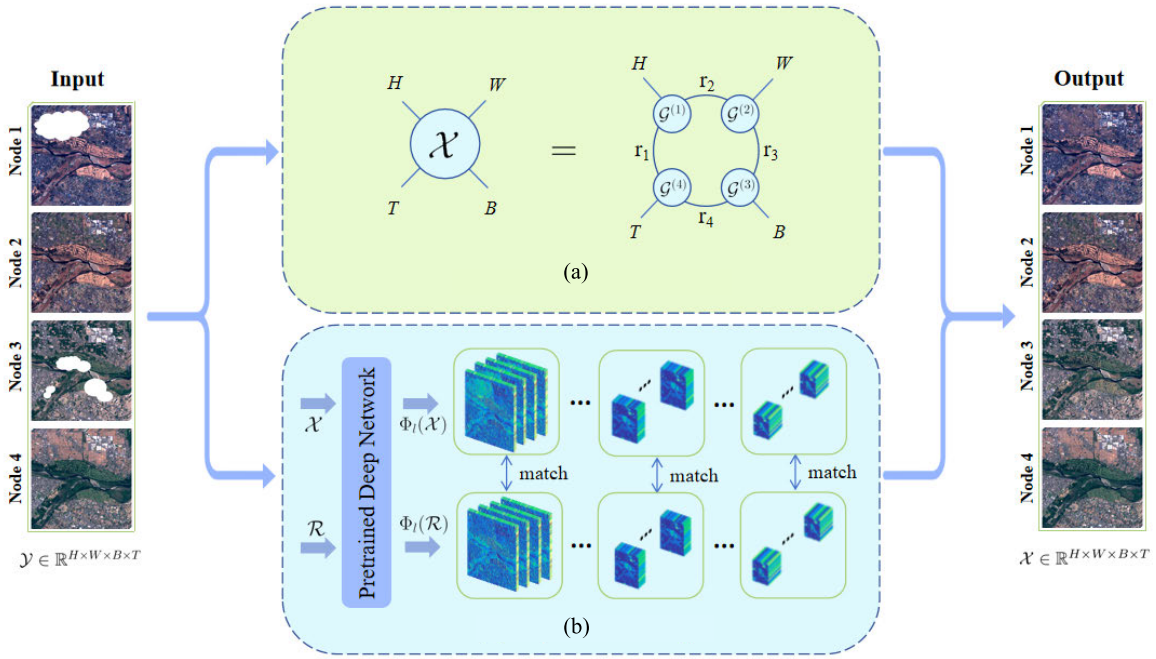


Fig. 3. Illustration of the proposed DFTR. With the observed image $\mathcal{Y}$ as input, (a) regularizes the reconstructed image $\mathcal{X}$ by TR decomposition. (b) Deep domain fidelity matches the deep features between $\mathcal{X}$ and guided image $\mathcal{R}$ of different levels by leveraging a pretrained deep network (i.e., AlexNet and VGG19).

ensuring model robustness. Using the previous model, such as the gradient domain, to reconstruct the image contaminated by cloud may not be a suitable fidelity for MTRS images because of its inadequacy in capturing invariant and delicate features.

Recently, deep learning (DL) has developed a significant advance in the environmental RS process [51], [52]. The deep convolutional neural networks (CNNs) are also gradually being employed in cloud removal tasks [53]. For example, Zhang et al. [54] filled gaps of cloud using spatiotemporal patch group DL. Singh and Komodakis [55] proposed a Cloud-GAN to remove the cloud of RS images. Chen et al. [56] removed the thick cloud from MTRS images using an LR regularized self-supervised network. Ebel et al. [57] proposed a multisensor time-series cloud removal method based on the residual network. Typically, through the complex network structure in DL, more potential features in RS images can be obtained with training data. And each layer of the pretrained deep CNNs contains the ability to capture deep features, which implies the essential information of the images. Compared with the traditional pixel-level method, the operation mode of convolution computation emphasizes the relationship between each pixel and its neighbors in every patch. Inspired by the powerful nonlinear representation of DL theory, the deep domain fidelity constructed by CNNs can complete deep semantic feature extraction and RS precision (i.e., detail) understanding.

To address the fact that features in the gradient domain are usually fragile and the lack of research related to fidelity terms, we propose a deep domain fidelity to extract the deep features with intrinsic information from MTRS images, which further complements the original pixel domain fidelity. In the process of deep domain fidelity, not only the invariant features (see example in Fig. 1.) of the image can be guaranteed but also the delicate features can be better obtained to achieve better reconstruction effects, particularly in preserving fine detail. The workflow diagram that illustrates the process in detail of the proposed DFTR is shown in Fig. 3.

The main contributions are summarized as follows.

1) We propose a deep domain fidelity to extract deep features of MTRS images, which more faithfully preserves details compared to traditional domain fidelity methods. This deep domain fidelity not only ensures the preservation of invariant features in overall structure reconstruction but also takes into account the delicate characteristics of the deep features behind MTRS images to preserve detail consistency.
2) In combination with deep domain fidelity and LR TR regularization, we suggest a thick cloud removal model (DFTR). The LR TR regularization captures the global structure of the MTRS image. Meanwhile, the deep domain fidelity focuses on preserving the intricate details. Extensive experimental results highlight the excellent performance of the DFTR model, which overcomes the shortcomings of traditional methods in detail preservation.

The remainder of this article proceeds as follows: Section II gives some basic definitions and preliminary notations. Section III proposes the DFTR model and solving algorithm. Section IV reports experimental results and gives discussions. Section V summarizes this article.

II. Notations and Preliminaries

A. Basic Definitions

In this section, we summarize some basic notations in Nomenclature.

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Definition 1 (First Mode-k Unfolding [58]): The first mode-k unfolding for an Nth-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ is denoted by $\mathbf{X}_{(k)} \in \mathbb{R}^{I_k \times I_1 \cdots I_{k-1} I_{k+1} \cdots I_N}$, where the element $\mathcal{X}(i_1, i_2, \ldots, i_N)$ of tensor $\mathcal{X}$ maps to $\mathbf{X}_{(k)}$ in position $(i_k, i_1 + \sum_{s=2, s\neq k}^{N}(i_s - 1)\prod_{m=1, m\neq k}^{s-1} I_m)$.

In addition, we use $\mathbf{X}_{(k)} = \text{Unfold}_k(\mathcal{X})$ to denote the first mode-k unfolding and its inverse operator is denoted as $\mathcal{X} = \text{Fold}_k(\mathbf{X}_{(k)})$.

Definition 2 (Second Mode-k Unfolding [58]): The second mode-k unfolding for an Nth-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ is denoted by $\mathbf{X}_{\langle k \rangle} \in \mathbb{R}^{I_k \times I_{k+1} \cdots I_N I_1 \cdots I_{k-1}}$, where the element $\mathcal{X}(i_1, i_2, \ldots, i_N)$ of tensor $\mathcal{X}$ maps to $\mathbf{X}_{<k>}$ in position (i_k, j) satisfies $j = i_k + \sum_{s=k+1}^{N}(i_s - 1)\prod_{p=k+1}^{s-1} I_p + \sum_{s=1}^{k-1}(i_s - 1)\prod_{p=1}^{s-1} I_p \prod_{q=k+1}^{N} I_q$.

B. TR Decomposition

Definition 3 (Multilinear Product [48]): The multilinear product of the core tensor $\mathcal{G}^{(k)} \in \mathbb{R}^{R_k \times I_k \times R_{k+1}}$ and $\mathcal{G}^{(k+1)} \in \mathbb{R}^{R_{k+1} \times I_{k+1} \times R_{k+2}}$ is denoted by $\mathcal{G}^{(k,k+1)} \in \mathbb{R}^{R_k \times I_k I_{k+1} \times R_{k+2}}$, where $k = 1, 2, \ldots, N$, and $R_1 = R_{N+1}$. And $\mathcal{G}^{(k,k+1)}(i_s + I_k(j_s - 1))$ is derived from $\mathcal{G}^{(k)}(i_s)\mathcal{G}^{(k+1)}(j_s)$, where $i_s = 1, 2, \ldots, I_k$ and $j_s = 1, 2, \ldots, I_{k+1}$.

Definition 4 (TR Decomposition [48]): The TR decomposition for an Nth-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ is denoted to decompose it into N third-order core tensors by the circular multilinear product. Taking the fourth-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3 \times I_4}$ as an example, the core tensor are $\mathcal{G} = \{\mathcal{G}^{(1)}, \mathcal{G}^{(2)}, \mathcal{G}^{(3)}, \mathcal{G}^{(4)}\}$, where $\mathcal{G}^{(n)} \in \mathbb{R}^{R_n \times I_n \times R_{n+1}}$, $n = 1, 2, 3, 4$, and $R_5 = R_1$. In addition, the TR rank is represented by the vector $\mathbf{r} = (R_1, R_2, R_3, R_4)$, and the tensor $\mathcal{X}$ element at position (i_1, i_2, i_3, i_4) can be represented as follows:

$$\mathcal{X}(i_1, i_2, i_3, i_4) = \text{Trace}\{\mathcal{G}^{(1)}(r_1, i_1, r_2)\mathcal{G}^{(2)}(r_2, i_2, r_3) \mathcal{G}^{(3)}(r_3, i_3, r_4)\mathcal{G}^{(4)}(r_4, i_4, r_1)\}.$$

By these definitions, for tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$, its TR decomposition can be denoted by $\mathcal{X} = \text{TR}(\mathcal{G})$.

III. Methodology

An MTRS image can be represented as $\mathcal{X} \in \mathbb{R}^{H \times W \times B \times T}$, where H and W denote the spatial resolutions, and B and T indicate the number of spectral bands and time nodes, respectively. The problem of cloud contamination in RS images has been a persistent concern. Due to the redundancy along the temporal dimension within the MTRS image, combinations of original domain fidelity and different LR regularizations (i.e., TR decomposition) have been successfully applied to cloud removal. However, the above methods focus only on the design of the regularization term while neglecting the design of the fidelity term, leading to unsatisfactory results. Driven by this lack of design, we propose a deep domain fidelity aimed at capturing the invariant and delicate features present in cloud-contaminated images, ensuring a more detailed reconstruction of the MTRS images.

A. Deep Domain Fidelity

Deep neural networks have shown their power in feature extraction, allowing the reconstruction of high-quality MTRS images is possible by extracting global deep features from pretrained networks. Inspired by this, instead of requiring the pixels of the reconstructed image $\mathcal{X}$ to exactly match those of the reference-guided image $\mathcal{R}$, we encourage them to have similar feature representations in the deep domain, with the pretrained network Φ can capture deep features itself.

Let $\Phi_l(\mathcal{X})$ be the feature of the first l layers ($l = 1, 2, \ldots, L$) of the network Φ when processing image $\mathcal{X}$, while also implying features of different levels. With the reference-guided image $\mathcal{R}$ estimated by the observed MTRS image, the different levels of deep domain fidelity can be expressed as follows:

$$\sum_{l=1}^{L} \alpha_l \|\Phi_l(\mathcal{X}) - \Phi_l(\mathcal{R})\|_F^2 \tag{1}$$

where α_l is the weight of loss for the first l layers.

Subsequently, we delineate the methodology for estimating the reference-guided image $\mathcal{R}$. Since MTRS images record the same geographical scene at different time nodes, similar information from various time nodes can be effectively utilized to assist in reconstruction. The reference-guided image $\mathcal{R}$ can be obtained by estimating the cloud-contaminated regions through cloud-free regions in the other temporal images.

Based on the strong temporal correlation between images at different time nodes as revealed by the time invariance in Fig. 1, and to address the issue of gradual changes due to illumination or other environmental factors, we use the following regression model.

First, for the observed image $\mathcal{Y} \in \mathbb{R}^{H \times W \times B \times T}$, assuming that the image on time node i_n ($n = 1, 2, \ldots, p$) is cloud-contaminated, and the image on time node j_m ($m = 1, 2, \ldots, q$) is cloud-free ($p + q = T$). By calculating the mean value of the cloud-free images as the average image $\bar{\mathcal{Y}} \in \mathbb{R}^{H \times W \times B}$, which can be formulated as follows:

$$\bar{\mathcal{Y}} = \frac{1}{q}\sum_{m=1}^{q} \mathcal{Y}^{j_m} \tag{2}$$

where $\mathcal{Y}^{j_m} \in \mathbb{R}^{H \times W \times B}$ is the j_mth cloud-free time node ($m = 1, 2, \ldots, q$) of the observed image $\mathcal{Y}$ and $\bar{\mathcal{Y}}$ is the mean value of the cloud-free images $\mathcal{Y}^{j_m}$ ($m = 1, 2, \ldots, q$).

Second, based on the relationship between the cloud-free regions of both the average image $\bar{\mathcal{Y}} \in \mathbb{R}^{H \times W \times B}$ and cloud-contaminated images, we can solve the least-squares problem to obtain the coefficients $\mathbf{W}_b^t$ and intercepts c_b^t

$$\min_{\mathbf{W}_b^t, c_b^t} \sum_{t=i_1}^{i_p}\sum_{b=1}^{B} \left\|\mathcal{P}_\Omega\left(\mathbf{Y}_b^t\right) - \left(\mathbf{W}_b^t \mathcal{P}_\Omega(\bar{\mathbf{Y}}_b) + c_b^t\right)\right\|_F^2 \tag{3}$$

where $\mathbf{Y}_b^t \in \mathbb{R}^{H \times W}$ is the bth spectral band and the tth time node of the observed image $\mathcal{Y}$, $\bar{\mathbf{Y}}_b \in \mathbb{R}^{H \times W}$ is the bth spectral band of the average image $\bar{\mathcal{Y}}$, Ω denotes the indicator set of cloud-free region locations, Ω^C denotes the indicator set of cloud-contaminated region locations, and $\mathcal{P}_\Omega(\cdot)$ is a projection operator that maps the pixels at Ω while setting all other pixels to 0.

On this basis, we can estimate the corrupt pixel of the guided image $\hat{\mathcal{Y}} \in \mathbb{R}^{H \times W \times B \times T}$ from the average image $\bar{\mathcal{Y}}$

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Algorithm 1 Process for Obtaining the Reference-Guided Image

Input: The observed MTRS image $\mathcal{Y}$, and the indicator set Ω.

Compute $\bar{\mathcal{Y}}$ via (2).
for $b = 1 : B$ **do**
  for $t = i_1, i_2, \ldots, i_p$ **do**
    Estimate $\mathbf{W}_b^t$ and c_b^t via (3).
    Estimate $\hat{\mathcal{Y}}$ via (4).
    Compute reference-guided image $\mathcal{R}$ via (5).
  end for
end for

Output: The reference-guided image $\mathcal{R}$.

band by band ($b = 1, 2, \ldots, B$)

$$\mathcal{P}_{\Omega^C}\left(\hat{\mathbf{Y}}_b^t\right) = \mathbf{W}_b^t \mathcal{P}_{\Omega^C}\left(\bar{\mathbf{Y}}_b\right) + c_b^t \quad \text{for } t = i_1, i_2, \ldots, i_p \tag{4}$$

where $\hat{\mathbf{Y}}_b^t \in \mathbb{R}^{H\times W}$ is the bth spectral band and tth time node of guided image $\hat{\mathcal{Y}}$.

Third, to obtain the reference-guided image $\mathcal{R} \in \mathbb{R}^{H\times W\times B\times T}$, we substitute the corresponding regions of the guided image $\hat{\mathcal{Y}}$ into the cloudy regions of the observed MTRS image $\mathcal{Y}$

$$\mathcal{R} = \mathcal{P}_{\Omega}(\mathcal{Y}) + \mathcal{P}_{\Omega^C}(\hat{\mathcal{Y}}). \tag{5}$$

The estimation process of reference-guided image is summarized in Algorithm 1.

B. Proposed DFTR Model and Optimization Algorithm

To recover the cloud-free MTRS image $\mathcal{X} \in \mathbb{R}^{H\times W\times B\times T}$ from the observation $\mathcal{Y} \in \mathbb{R}^{H\times W\times B\times T}$ that is cloud-contaminated, we use deep domain fidelity to capture both invariant features and delicate features, thereby enhancing the reconstruction of details in the cloud-contaminated image. Based on the deep domain fidelity, we suggest a DFTR model as follows:

$$\begin{aligned} &\min_{\mathcal{X},\mathcal{G}} \ \frac{1}{2}\|\mathcal{X} - \mathrm{TR}(\mathcal{G})\|_F^2 + \sum_{l=1}^{L}\frac{\alpha_l}{2}\|\Phi_l(\mathcal{X}) - \Phi_l(\mathcal{R})\|_F^2 \\ &\text{s.t.} \ \ \mathcal{P}_{\Omega}(\mathcal{X}) = \mathcal{P}_{\Omega}(\mathcal{Y}) \end{aligned} \tag{6}$$

where Φ_l is the first l layers of network Φ and α_l is the weight of loss for the first l layers.

According to the framework of alternating minimization algorithm [59], which alternately updates one variable while keeping the others fixed, the problem (6) can be solved by iteratively updating each variable as follows:

$$\begin{cases} \mathcal{G}^{(1),k+1} = \arg\min_{\mathcal{G}^{(1)}} f\left(\mathcal{G}^{(1)}, \mathcal{G}^{(2),k}, \mathcal{G}^{(3),k}, \mathcal{G}^{(4),k}, \mathcal{X}^k\right) \\ \mathcal{G}^{(2),k+1} = \arg\min_{\mathcal{G}^{(2)}} f\left(\mathcal{G}^{(1),k+1}, \mathcal{G}^{(2)}, \mathcal{G}^{(3),k}, \mathcal{G}^{(4),k}, \mathcal{X}^k\right) \\ \mathcal{G}^{(3),k+1} = \arg\min_{\mathcal{G}^{(3)}} f\left(\mathcal{G}^{(1),k+1}, \mathcal{G}^{(2),k+1}, \mathcal{G}^{(3)}, \mathcal{G}^{(4),k}, \mathcal{X}^k\right) \\ \mathcal{G}^{(4),k+1} = \arg\min_{\mathcal{G}^{(4)}} f\left(\mathcal{G}^{(1),k+1}, \mathcal{G}^{(2),k+1}, \mathcal{G}^{(3),k+1}, \mathcal{G}^{(4)}, \mathcal{X}^k\right) \\ \mathcal{X}^{k+1} = \arg\min_{\mathcal{X}} f\left(\mathcal{G}^{k+1}, \mathcal{X}\right) \end{cases} \tag{7}$$

Algorithm 2 Proposed DFTR Model for Thick Cloud Removal

Input: The observed MTRS image $\mathcal{Y}$, the TR rank $\mathbf{r}$, the weight α_l, the penalty parameters β, and the indicator set Ω.

Initialization: $\mathcal{X}^{(0)} = \mathcal{Y}$, $k = 0$, and $k_{\max} = 10000$.

Estimate the reference-guided image $\mathcal{R}$ by Algorithm 1.
while not converged and $k < k_{\max}$ **do**
  for $n = 1, 2, 3, 4$ **do**
    Updating $\mathcal{G}^{(n),k+1}$ via (10).
  end for
  Updating $\mathcal{X}^{k+1}$ using the Adam algorithm.
  Let $k = k + 1$.
end while
Compute $\mathcal{X} = \mathcal{P}_{\Omega}(\mathcal{Y}) + \mathcal{P}_{\Omega^C}(\mathcal{X})$.

Output: The reconstructed MTRS image $\mathcal{X}$.

where $\mathcal{G}^{k+1} = \{\mathcal{G}^{(1),k+1}, \mathcal{G}^{(2),k+1}, \mathcal{G}^{(3),k+1}, \mathcal{G}^{(4),k+1}\}$ and $f(\mathcal{G}, \mathcal{X})$ is the composite of the objective function and constraints of (6).

1) Updating $\mathcal{G}^{(n)}$: The core tensor $\mathcal{G}^{(n)}$ subproblem is as follows:

$$\min_{\mathcal{G}^{(n)}} \ \frac{1}{2}\left\|\mathcal{X}^k - \mathrm{TR}\left(\mathcal{G}^{(1:n-1),k+1}, \mathcal{G}^{(n)}, \mathcal{G}^{(n+1:N),k}\right)\right\|_F^2. \tag{8}$$

According to Theorem 3.5 of the TR-ALS algorithm in [48], it can be transformed into the following formulation:

$$\min_{\mathcal{G}^{(n)}} \ \frac{1}{2}\left\|\mathcal{X}_{<n>}^k - \mathbf{G}_{(2)}^{(n)}\left(\mathbf{G}_{<2>}^{(\neq n),k}\right)^{\mathrm{T}}\right\|_F^2. \tag{9}$$

The closed-form solution of the least-squares problem (9) is as follows:

$$\mathcal{G}^{(n),k+1} = \mathrm{Fold}_2\left(\left(\mathbf{X}_{\langle n\rangle}^k \mathbf{G}_{\langle 2\rangle}^{(\neq n),k}\right) \times \left(\left(\mathbf{G}_{\langle 2\rangle}^{(\neq n),k}\right)^{\mathrm{T}} \mathbf{G}_{\langle 2\rangle}^{(\neq n),k}\right)^{-1}\right) \tag{10}$$

where $n = 1, 2, 3, 4$.

2) Updating $\mathcal{X}$: The $\mathcal{X}$ subproblem is as follows:

$$\begin{aligned} \min_{\mathcal{X}} \ &\frac{1}{2}\left\|\mathcal{X} - \mathrm{TR}\left(\mathcal{G}^{k+1}\right)\right\|_F^2 \\ &+ \sum_{l=1}^{L}\frac{\alpha_l}{2}\|\Phi_l(\mathcal{X}) - \Phi_l(\mathcal{R})\|_F^2 + \frac{\beta}{2}\psi_{\mathbb{A}}(\mathcal{X}) \end{aligned} \tag{11}$$

where

$$\psi_{\mathbb{A}}(\mathcal{X}) := \begin{cases} 0, & \text{if } \mathcal{X} \in \mathbb{A} \\ \infty, & \text{otherwise} \end{cases} \tag{12}$$

with $\mathbb{A} := \{\mathcal{X} : \mathcal{P}_{\Omega}(\mathcal{X}) = \mathcal{P}_{\Omega}(\mathcal{Y})\}$ and β is the penalty parameter. This subproblem (11) is a highly nonconvex and nonlinear optimization problem, which we can solve efficiently using Adam [60].

The solving algorithm of the proposed DFTR model is summarized in Algorithm 2.

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TABLE I
COMPREHENSIVE DETAILS ON MTRS IMAGES FOR THE FOUR SIMULATED DATASETS

Dataset	Source	Time nodes	Spectral bands	Image size	Spatial resolution
Tokyo	Sentinel-2	4	4	400×400	10 m
Picardie	Sentinel-2	6	6	500×500	20 m
France	Sentinel-2	4	6	600×600	10 m
Bourgogne	Landsat 8	6	7	400×400	30 m

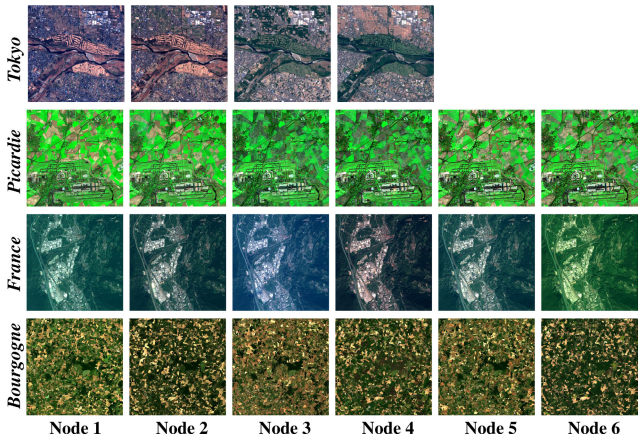


Fig. 4. Visual display of MTRS images for the simulated datasets.

IV. NUMERICAL EXPERIMENTS

This section tests the thick cloud removal performance of the proposed DFTR for MTRS images on both simulated and real data. To demonstrate the superiority of the proposed method, we compare it with six methods, including Regression, HaLRTC [38], TNN [61], TT [46], TVTR [49], and TRGFid [50]. Regression employs the least-squares method to estimate the reconstructed images at cloud-contaminated time nodes based on remaining images. Reconstruction methods such as HaLRTC, TNN, and TT utilize different LR regularizations with original domain fidelity. Methods like TVTR and TRGFid are based on TR decomposition, with TVTR incorporating TV regularization and TRGFid combining gradient-domain fidelity. The parameters of each comparison method are manually fine-tuned for optimal performance based on the authors' code and recommendations. The proposed method is implemented through the PyTorch 2.1.0 based on NVIDIA RTX 4060Ti GPU with 16-GB RAM. The experiments of the comparison method are performed on MATLAB R2023a based on an Intel Core i7-12700 CPU with 32-GB RAM.

A. Simulated Experiments

We conduct simulated experiments to test the performance of the proposed DFTR using MTRS images from Sentinel-2[1] and Landsat 8.[2] All experimental data presented in this article are sourced from publicly available datasets. More details and information related to the test datasets and the evaluation indices are provided below.

[1]https://earthexplorer.usgs.gov
[2]https://theia.cnes.fr/atdistrib/rocket/#/home

1) Datasets: To test the effectiveness and robustness of DFTR, we use four datasets to perform simulated experiments. The datasets for *Tokyo*, *Picardie*, and *France* are taken by Sentinel-2 over Tokyo, Picardie, and France, respectively. The dataset for *Bourgogne* is taken by Landsat 8 over Bourgogne. Table I and Fig. 4 show the details of the datasets used for the simulated experiment, including the number of time nodes and spectral bands, spatial resolution, and image size.

We add different simulated clouds to each dataset in experiments. Two time nodes in the *Tokyo* dataset are chosen to add simulated clouds. Three time nodes in each of the *France*, *Picardie*, and *Bourgogne* datasets are chosen to add simulated clouds. In addition, we assume that the clouds appear at different locations at different time nodes, with varying sizes and shapes. To ensure fairness, the same reference images were used in both the comparison method and the proposed method experiments. However, it is noted that the simulated cloud occupies the same location across all bands at a given time node.

2) Evaluation Indices: We adopt three image quality metrics as evaluation indices to objectively assess the cloud removal methods through numerical comparisons. The peak signal-to-noise ratio (PSNR) can effectively reflect the visual reconstruction quality of the image, with higher PSNR values indicating better reconstruction performance

$$\mathrm{PSNR} = \frac{1}{B}\sum_{b=1}^{B} 10 \times \log_{10} \frac{HW\left(\tilde{\mathbf{X}}_b\right)_{\max}^2}{\left\|\mathbf{X}_b - \tilde{\mathbf{X}}_b\right\|_F^2}$$

where $\tilde{\mathcal{X}}$ represents the ground truth and $(\tilde{\mathbf{X}}_b)_{\max}$ indicates the maximum pixel value in $\tilde{\mathbf{X}}_b$.

The structural similarity (SSIM) measures the reconstruction quality with higher values being better from three different perspectives, including brightness, contrast, and structure of the image

$$\mathrm{SSIM} = \frac{1}{B}\sum_{b=1}^{B} \frac{\left(2\mu_{\mathbf{X}_b}\mu_{\tilde{\mathbf{X}}_b} + c_1\right)\left(2\sigma_{\mathbf{X}_b\tilde{\mathbf{X}}_b} + c_2\right)}{\left(\mu_{\mathbf{X}_b}^2 + \mu_{\tilde{\mathbf{X}}_b}^2 + c_1\right)\left(\sigma_{\mathbf{X}_b}^2 + \sigma_{\tilde{\mathbf{X}}_b}^2 + c_2\right)}$$

where $\mu_{\mathbf{X}_b}$ and $\mu_{\tilde{\mathbf{X}}_b}$ are the average values of $\mathbf{X}_b$ and $\tilde{\mathbf{X}}_b$, respectively; $\sigma_{\mathbf{X}_b\tilde{\mathbf{X}}_b}$ is the covariance between $\mathbf{X}_b$ and $\tilde{\mathbf{X}}_b$; $\sigma_{\mathbf{X}_b}$ and $\sigma_{\tilde{\mathbf{X}}_b}$ are the standard deviation of $\mathbf{X}_b$ and $\tilde{\mathbf{X}}_b$, respectively; and c_1, c_2 are constants.

Spectral angle mapping (SAM), which measures the similarity between images by comparing the angles between their spectral vectors, indicates higher similarity to the original image with a smaller SAM value

$$\mathrm{SAM} = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W} \cos^{-1}\left(\frac{\left(\mathbf{X}_{i,j}\right)^{\mathrm{T}}\tilde{\mathbf{X}}_{i,j}}{\left\|\mathbf{X}_{i,j}\right\|_F\left\|\tilde{\mathbf{X}}_{i,j}\right\|_F + c_3}\right)$$

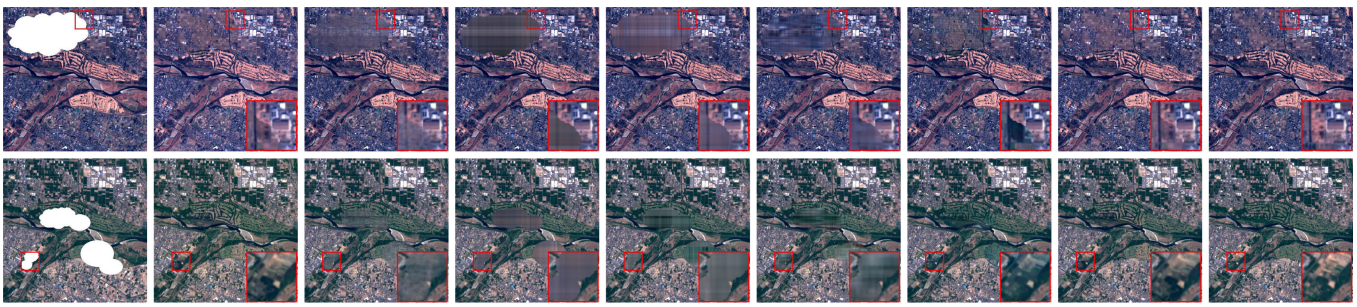
where $\mathbf{X}_{i,j}$ and $\tilde{\mathbf{X}}_{i,j}$ denote the columns of $\mathcal{X}$ and $\tilde{\mathcal{X}}$ corresponding to fixed indices i and j, respectively; and c_3 is a constant.

3) Quantitative Comparison: To analyze the performance of DFTR quantitatively, Table II presents the PSNR, SSIM, and SAM values of the reconstructed images obtained using various cloud removal methods. Bold text and underlined text indicate the best and second-best results, respectively. It can

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TABLE II
QUANTITATIVE EVALUATION RESULTS BASED ON DIFFERENT METHODS UNDER DIFFERENT TIME NODES

Dataset	Contaminated nodes	Index	Observed	Regression	HaLRTC	TNN	TT	TVTR	TRGFid	DFTR
Tokyo	node 1	PSNR	11.148	37.240	37.587	35.733	36.630	36.631	37.704	**40.443**
		SSIM	0.8823	0.9742	0.9571	0.9505	0.9517	0.9519	0.9598	**0.9751**
		SAM	1.3041	1.9244	0.5778	0.9428	0.6912	0.6361	0.5559	**0.4465**
	node 2	PSNR	12.706	39.780	37.575	36.017	36.764	37.815	40.106	**41.803**
		SSIM	0.8961	0.9842	0.9669	0.9582	0.9587	0.9632	0.9802	**0.9850**
		SAM	2.5850	1.3729	0.7594	1.0793	0.7952	0.6234	0.4547	**0.3981**
		Time(s)	—	**0.16**	65.23	88.21	49.59	45.42	132.19	340.27
Picardie	node 1	PSNR	7.561	44.531	42.124	43.096	40.142	41.970	45.642	**46.765**
		SSIM	0.7938	0.9914	0.9785	0.9763	0.9644	0.9714	0.9930	**0.9932**
		SAM	2.1447	2.4364	1.0995	1.2313	1.3952	1.4802	0.8263	**0.7033**
	node 2	PSNR	4.328	44.817	40.674	40.523	38.644	40.199	44.940	**45.121**
		SSIM	0.5827	0.9883	0.9702	0.9642	0.9496	0.9608	0.9858	**0.9885**
		SAM	5.3596	1.5428	2.1666	2.7682	3.1202	2.4294	1.4846	**1.2623**
	node 3	PSNR	9.539	46.618	48.386	46.285	46.395	49.031	49.651	**51.612**
		SSIM	0.8611	0.9952	0.9946	0.9918	0.9909	0.9932	0.9959	**0.9976**
		SAM	1.4347	1.3690	0.3614	0.6192	0.6609	0.4787	0.3881	**0.2446**
		Time(s)	—	**0.37**	202.73	160.07	59.53	247.07	384.40	348.75
France	node 1	PSNR	7.559	47.791	44.561	46.885	41.363	42.546	47.860	**48.965**
		SSIM	0.7979	0.9947	0.9871	0.9880	0.9770	0.9756	0.9947	**0.9950**
		SAM	3.5046	1.3532	0.7990	0.7163	1.2426	1.0487	0.7070	**0.3942**
	node 2	PSNR	9.976	45.576	47.059	49.333	44.387	44.856	49.172	**51.874**
		SSIM	0.8770	0.9955	0.9923	0.9924	0.9867	0.9845	0.9962	**0.9972**
		SAM	2.0942	2.5094	0.4620	0.4247	0.6267	0.6023	0.3680	**0.2265**
	node 3	PSNR	6.052	45.556	42.966	39.766	38.280	41.684	45.653	**46.856**
		SSIM	0.7185	0.9930	0.9843	0.9848	0.9719	0.9734	0.9932	**0.9936**
		SAM	4.7017	0.7512	1.0437	0.8963	1.7166	1.2382	0.8306	**0.5489**
		Time(s)	—	**0.28**	367.70	186.61	27.37	216.24	270.53	389.57
Bourgogne	node 1	PSNR	15.141	26.981	40.877	40.426	40.527	40.789	40.434	**42.977**
		SSIM	0.9466	0.9294	0.9788	0.9798	0.9782	0.9791	0.9835	**0.9868**
		SAM	1.9657	3.6567	0.4547	0.4602	0.4699	0.4580	0.3980	**0.3774**
	node 2	PSNR	8.858	31.463	30.000	30.452	29.746	30.070	31.480	**31.908**
		SSIM	0.7751	0.9353	0.8814	0.8952	0.8787	0.8823	0.9346	**0.9369**
		SAM	8.4220	3.0716	2.6429	2.4955	2.6898	2.5648	2.0600	**1.9279**
	node 3	PSNR	12.227	27.718	34.597	34.547	34.228	34.639	34.525	**35.488**
		SSIM	0.8976	0.9157	0.9454	0.9498	0.9445	0.9463	0.9540	**0.9604**
		SAM	3.8395	4.2123	0.9520	1.0079	1.0360	0.9546	1.0931	**0.9457**
		Time(s)	—	**0.22**	141.98	144.88	10.70	93.13	162.37	786.46
	Average	PSNR	9.554	37.666	40.218	40.383	38.896	39.939	42.055	**43.932**
		SSIM	0.8208	0.9694	0.9732	0.9735	0.9707	0.9739	0.9816	**0.9847**
		SAM	3.3960	2.2461	1.0977	1.1499	1.2272	1.0971	0.7399	**0.5383**
		Time(s)	—	**0.26**	194.41	144.94	36.80	150.47	237.37	466.76



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 5. Experimental results from various cloud removal methods of the *Tokyo* dataset. The images (R: B3, G: B2, and B: B1) are presented from top to bottom at contaminated nodes 1 and 2. (a) Observed. (b) Regression. (c) HaLRTC. (d) TNN. (e) TT. (f) TVTR. (g) TRGFid. (h) DFTR. (i) Ground truth.

be observed in Table II that the proposed method DFTR outperforms other compared methods across all of the datasets. Among these cloud removal methods, TNN and TRGFid perform well with most of the datasets. In comparison, the

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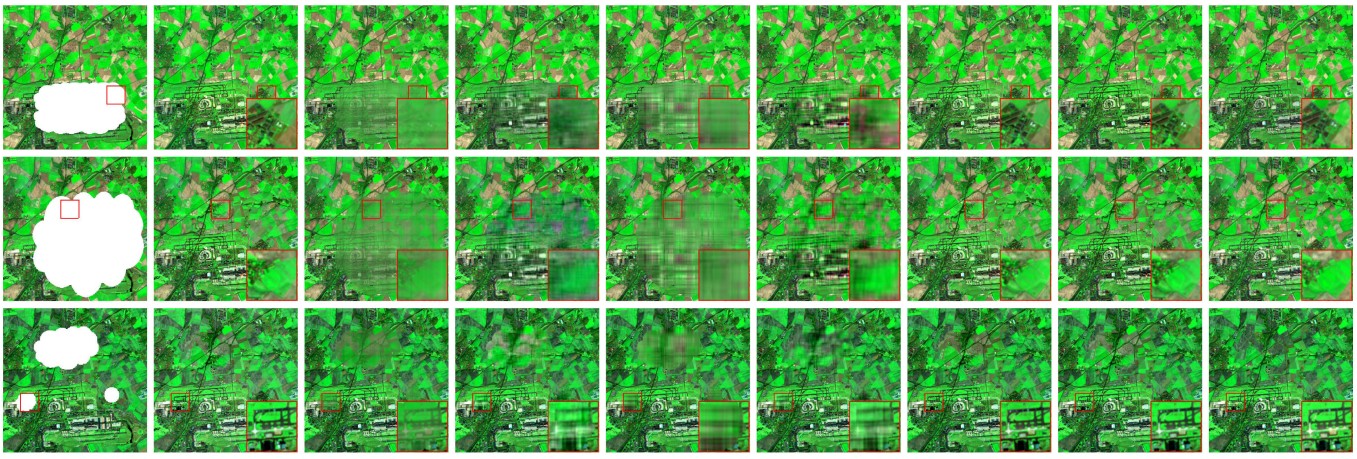


Fig. 6. Experimental results from various cloud removal methods of the *Picardie* dataset. The images (R: B6, G: B4, and B: B1) are presented from top to bottom at contaminated nodes 1 and 2. (a) Observed. (b) Regression. (c) HaLRTC. (d) TNN. (e) TT. (f) TVTR. (g) TRGFid. (h) DFTR. (i) Ground truth.

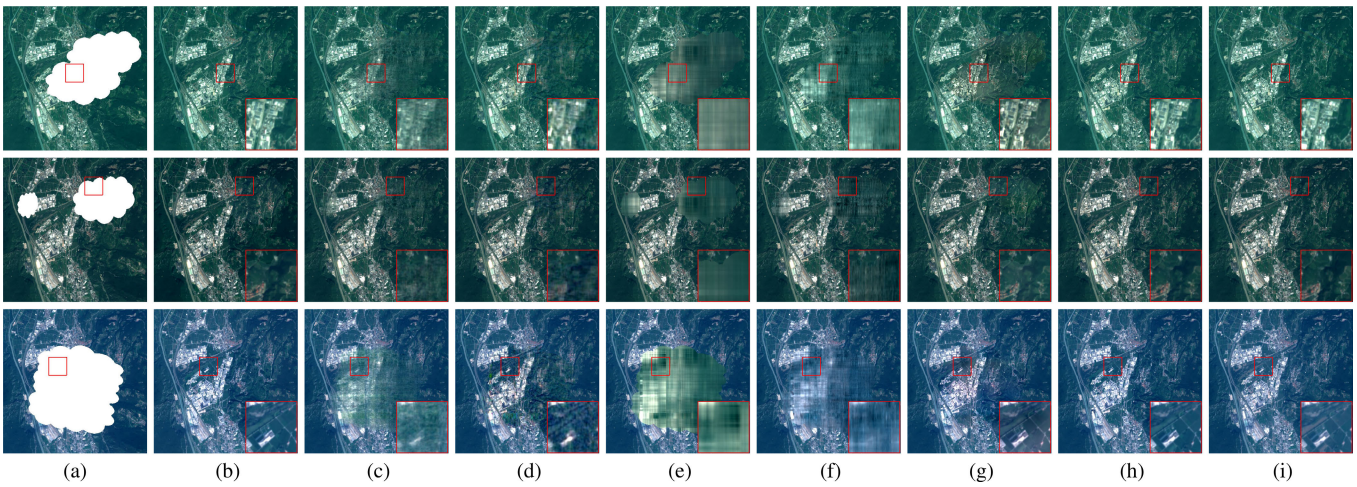


Fig. 7. Experimental results from various cloud removal methods of the *France* dataset. The images (R: B3, G: B2, and B: B1) are presented from top to bottom at contaminated nodes 1 and 2. (a) Observed. (b) Regression. (c) HaLRTC. (d) TNN. (e) TT. (f) TVTR. (g) TRGFid. (h) DFTR. (i) Ground truth.

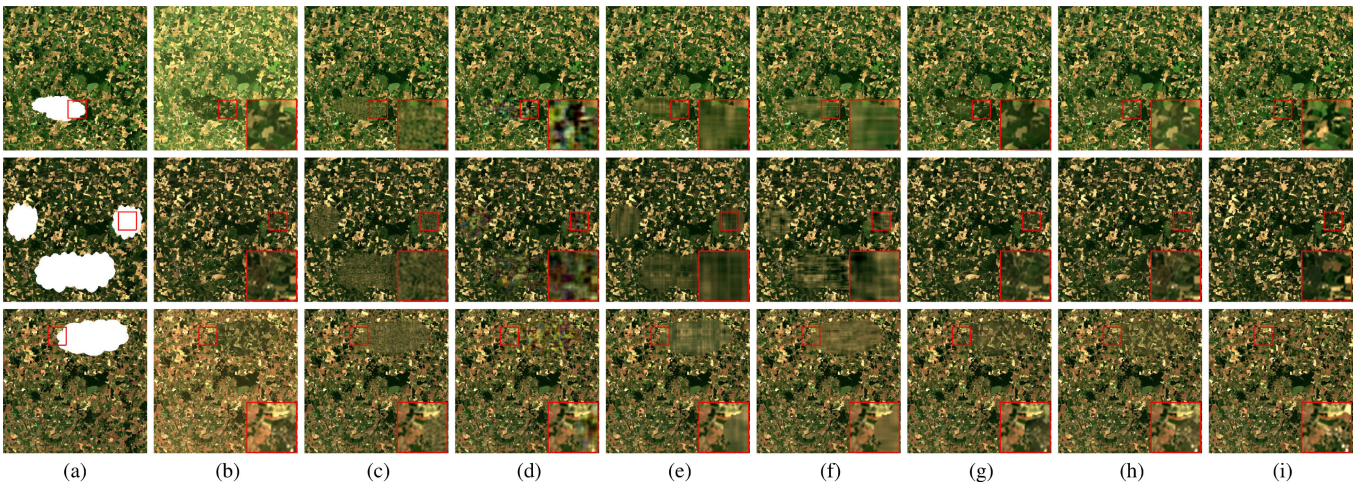


Fig. 8. Experimental results from various cloud removal methods of the *Bourgogne* dataset. The images (R: B4, G: B3, and B: B2) are presented from top to bottom at contaminated nodes 1 and 2. (a) Observed. (b) Regression. (c) HaLRTC. (d) TNN. (e) TT. (f) TVTR. (g) TRGFid. (h) DFTR. (i) Ground truth.

proposed DFTR can achieve better results because deep domain fidelity has a significant impact on detail preservation. Although the running time of the proposed method is higher than the comparison methods, the proposed method consistently delivers superior results within an acceptable time. It is worth noting that the proposed DFTR achieves

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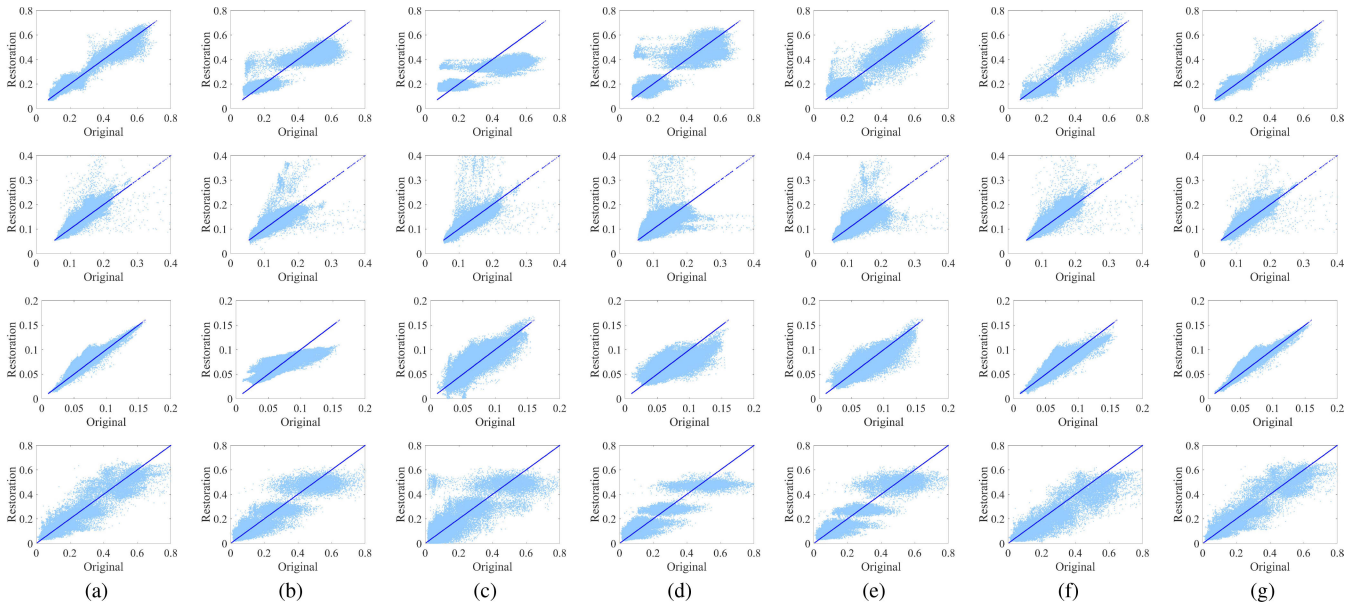


Fig. 9. Scatter plots comparing the original and reconstruction pixels in the missing areas for different simulated datasets. The datasets are shown from top to bottom as follows: *Tokyo* at contaminated nodes 2, *France* at contaminated nodes 3, *Picardie* at contaminated nodes 3, and *Bourgogne* at contaminated nodes 1. (a) Regression. (b) HaLRTC. (c) TNN. (d) TT. (e) TVTR. (f) TRGFid. (g) DFTR.

TABLE III
COMPREHENSIVE DETAILS ON MTRS IMAGES FOR THE TWO REAL DATASETS

Dataset	Source	Time nodes	Spectral bands	Image size	Spatial resolution
Bourgogne	Landsat 8	4	7	600×600	30 m
Beijing	SPOT-5	4	4	1000×1000	10 m

around 2-dB gain in average PSNR beyond the second-best method, benefiting from the fact that the proposed method fully exploits the invariant and delicate information within the spatial–spectral–temporal structure of the MTRS images.

4) Qualitative Comparison: To qualitatively analyze the performance of DFTR, the reconstructed images for different cloud removal methods and ground truth are shown in Figs. 5–8. The locations indicated by red rectangles in each image are magnified for enhanced visualization and comparison. As observed in Figs. 5–8, the proposed method DFTR is effective at restoring detail in the reconstruction results. The comparison indicates that the reconstruction results obtained by TRGFid generally have more edge and texture information than those obtained by other methods except DFTR. While DFTR ensures the recovery of the overall structure, it also reduces detail deviation effectively. This is because deep domain fidelity not only maintains the consistent and invariant features of the RS image but also more comprehensively deconstructs the RS image through the understanding of the delicate relationship between neighboring pixels, which ensures the fine detail of the RS image.

To further evaluate the quality of the reconstruction, Fig. 9 shows the scatter plots between the original pixels and the reconstructed pixels in the cloud-contaminated regions of all simulated datasets by each method. The closer the overall distribution of points is to the diagonal line, the more similar the reconstructed pixel points are to the original pixel points.

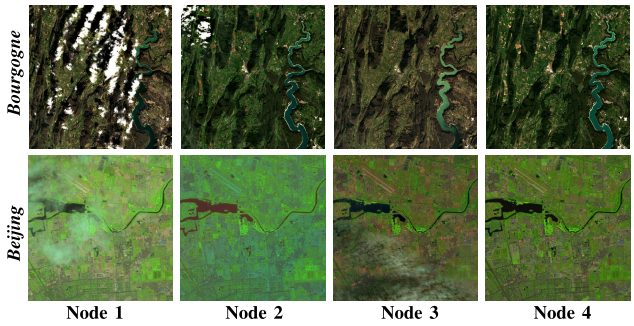


Fig. 10. Visual display of MTRS images for the real datasets.

It can be observed that the reconstructed pixel points of our proposed method are closest to the original pixel points, which intuitively proves the superiority of DFTR. Many points of HaLRTC, TNN, and TT deviate from the diagonal, indicating that the reconstruction is not satisfactory. The distributions of points in TVTR and TRGFid are more concentrated, but there are still some points that deviate from the original pixel point. The scatter diagram distribution of Regression is similar to that of DFTR, which illustrates its effectiveness in providing auxiliary reference information for DFTR. This observation corroborates with the visualization presented in Figs. 5–8 and the quantitative analysis presented in Table II, further demonstrating the superiority of the DFTR method. This comprehensive method leads to more accurate and visually appealing reconstructions.

B. Real Experiments

In the real experiments, two datasets, *Bourgogne* and *Beijing*, are selected to evaluate the performance of the proposed method in real scenarios. Table III provides the details of these datasets, and Fig. 10 displays their respective images. The *Bourgogne* images, taken by Landsat 8 over

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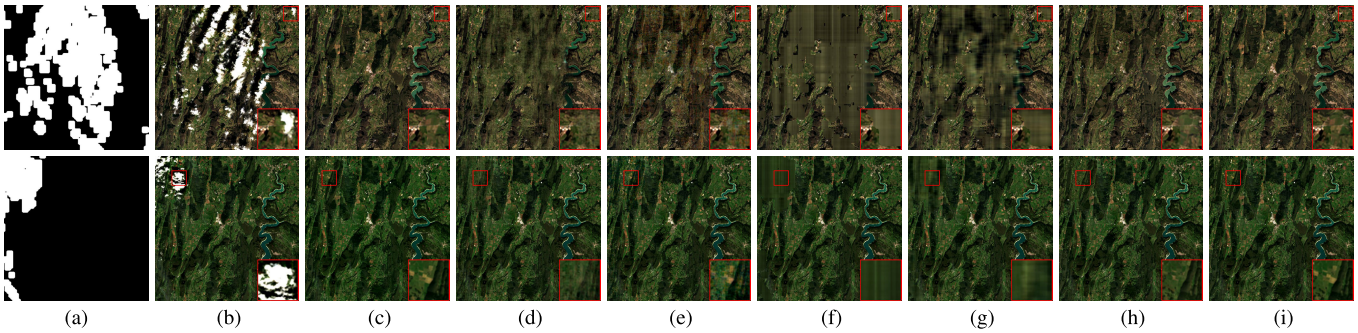


Fig. 11. Experimental results from various cloud removal methods of the real *Bourgogne* dataset. The images (R: B4, G: B3, and B: B2) are presented from top to bottom at contaminated nodes 1 and 2. (a) Mask. (b) Observed. (c) Regression. (d) HaLRTC. (e) TNN. (f) TT. (g) TVTR. (h) TRGFid. (i) DFTR.

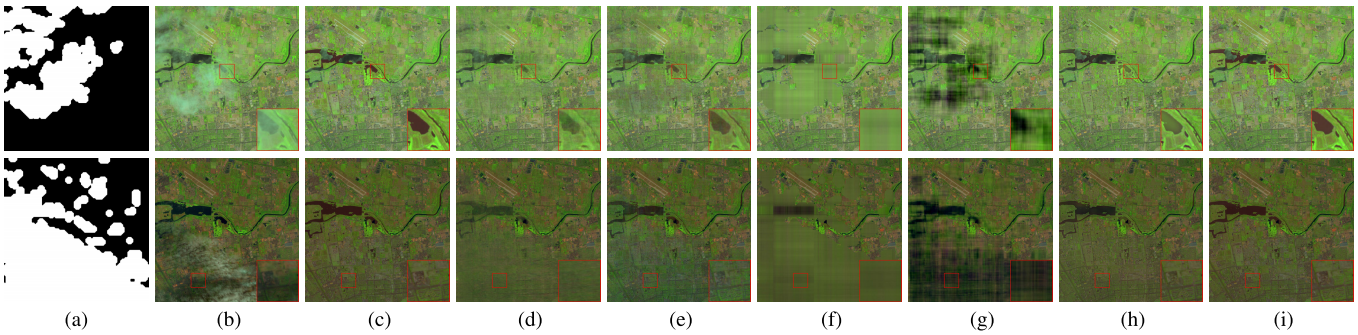


Fig. 12. Experimental results from various cloud removal methods of the real *Beijing* dataset. The images (R: B4, G: B3, and B: B2) are presented from top to bottom at contaminated nodes 1 and 2. (a) Mask. (b) Observed. (c) Regression. (d) HaLRTC. (e) TNN. (f) TT. (g) TVTR. (h) TRGFid. (i) DFTR.

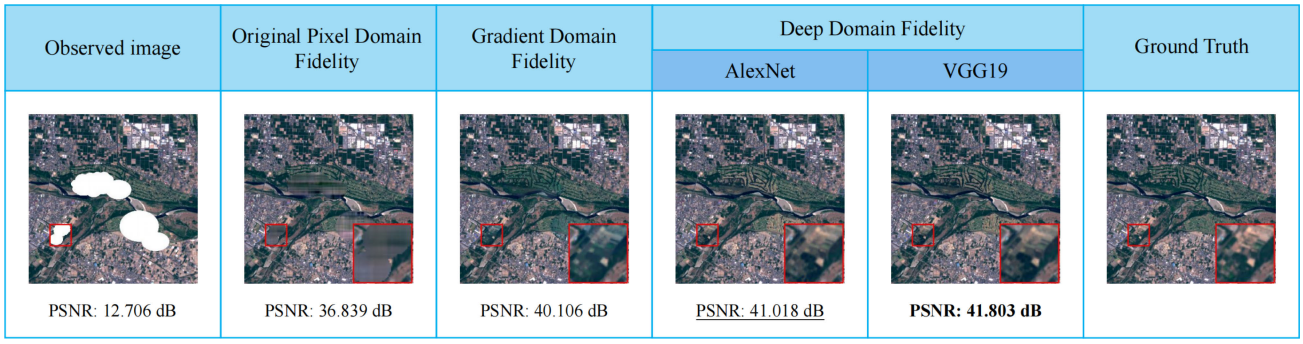


Fig. 13. Role of different domain fidelities on the *Tokyo* dataset.

Bourgogne, exhibit the presence of cloud contamination at the first two time nodes, with the cloud masks accessible via the website.[2] The *Beijing* images, taken by SPOT-5[3] over Beijing, exhibit the presence of cloud contamination at the first time node and the third time node, with the cloud masks accessible via the website.[4] The remaining clean images are employed as supplementary information for the estimation of the reference-guided images.

To facilitate visual comparison, Figs. 11 and 12 show the reconstruction results of all cloud removal methods. From the observation, the TT and TVTR methods are inadequate for the reconstruction of details. While HaLRTC and TNN are capable of reconstructing the overall structure, the texture remains deficient. The reconstructions achieved by Regression and TRGFid exhibit better textures and edges, but the preservation of details is still unsatisfactory. In comparison, the proposed DFTR outperforms the other methods in terms of reconstruction quality. This is because deep domain fidelity focuses on both invariant and delicate features of the RS image during cloud removal.

C. Discussion

1) Role of Different Domain Fidelity: This section provides a comparative analysis of the roles of different domain fidelities on cloud removal performance. To comprehensively measure the impact of domain fidelity on the reconstruction quality, we present the reconstruction results of different domain fidelities and show them in Fig. 13.

The comparison shows that both gradient and deep domain fidelity outperform the original pixel domain fidelity in reconstructing RS images, with significant improvements in texture and edge preservation. Furthermore, comparing data-driven

[3]https://take5.theia.cnes.fr/atdistrib/take5/client/#/home
[4]http://www.cesbio.ups-tlse.fr/multitemp/?p=6203

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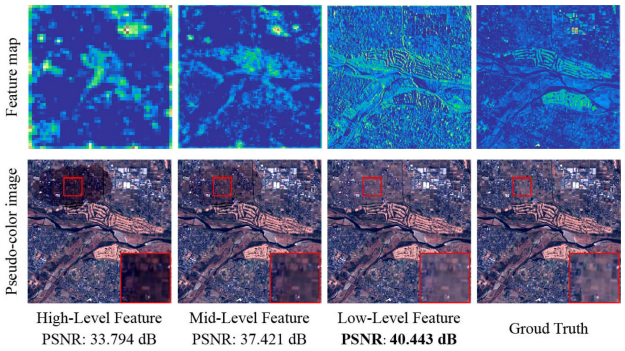


Fig. 14. Effect of different levels of feature on the *Tokyo* dataset. The first row is the feature maps at the high level, mid-level, and low level, respectively, which have been averaged along the band dimension. The second row is the corresponding reconstruction results.

deep domain fidelity to handcrafted gradient-domain fidelity, we can observe that gradient-domain fidelity may not be a suitable fidelity for MTRS images because of its detail inaccuracy. Whereas deep domain fidelity can capture more intrinsic deep features, ensuring better consistency of detail.

More concretely, we compare two pretrained networks, AlexNet and VGG19, which differ in the number of network layers and the progression between each layer, and VGG19 provides better reconstruction results. Compared to AlexNet, VGG19 has more network layers, deeper depth, and a more elaborate network structure design. These allow the network VGG19 to better access the more essential features of images embedded in its pretrained network. Unlike AlexNet, which uses 11×11 convolution kernel in the first layer to extract coarser features, VGG19 uses 3×3 convolution kernel, which helps VGG19 to acquire more details in image reconstruction.

2) Discussion of the Levels of Features: This section discusses the influence of different levels of the deep features extracted by pretrained networks in deep domain fidelity. Deep domain fidelity grasping features can be classified into a different level of features by the depth of the pretrained network used. Taking VGG19 as an example, which has a total of 37 network layers, its first one-layer network, 11-layer network, and 21-layer network reflect low-level, mid-level, and high-level features, respectively. Fig. 14 shows the visual results of the feature maps and their corresponding reconstructed images for the three levels.

From observation, low-level features are beneficial for pixel-level reconstruction tasks, whereas mid-level and high-level feature maps, which capture more abstract features, do not offer adequate support for pixel-level reconstruction. The mid-level features and high-level features are critical for understanding the context and meaning of the image content, enabling tasks such as classification, detection, and segmentation. However, the RS image reconstruction performances of these features are not desirable. On the contrary, the low-level features are crucial for identifying basic structures in the image. They are typically less abstract and more related to pixel-level information.

3) Influence of the Cloud Coverage Ratio: This section discusses the influence of cloud contamination of different sizes on the effect of the proposed cloud removal method.

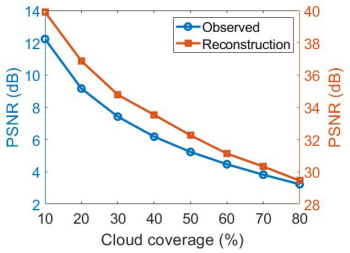


Fig. 15. Influence of the cloud coverage ratio on the reconstruction effect.

TABLE IV
QUANTITATIVE COMPARISON FOR THE EFFECT OF THE NUMBER OF REFERENCE IMAGES ON THE PROPOSED METHODS

Contaminated nodes	Index	Observed	Number of reference images		
			one	two	three
node 1	PSNR	15.141	41.880	41.977	**42.977**
	SSIM	0.9466	0.9842	0.9847	**0.9868**
	SAM	1.9657	0.4182	0.4114	**0.3774**
node 2	PSNR	8.858	30.716	31.410	**31.908**
	SSIM	0.7751	0.9231	0.9337	**0.9369**
	SAM	8.4220	2.3448	2.0802	**1.9279**
node 3	PSNR	12.227	34.729	34.876	**35.488**
	SSIM	0.8976	0.9527	0.9544	**0.9604**
	SAM	3.8395	1.1040	1.0858	**0.9457**
Average	PSNR	12.075	35.775	36.088	**36.791**
	SSIM	0.8731	0.9533	0.9585	**0.9612**
	SAM	4.7424	1.2890	1.1925	**1.0834**

We experimented by replacing masks with different cloud coverage (e.g., 10%, 20%, . . . , 80%) at the first time node in the *Tokyo* dataset. It can be intuitively observed from Fig. 15 that the PSNR value of the reconstructed image shows a decreasing trend with the increase of cloud coverage. More details of the reconstructed image for different cloud coverage can be seen in Fig. 16.

4) Effect of the Reference Images: This section discusses the effect of the number of reference images on the quality and effectiveness of RS image reconstruction. When the time series is short, the reliance on a single reference image can lead to biased reconstructions. The limited data availability means that any anomalies or unique characteristics in the single reference image heavily influence the reconstruction process.

The reconstruction of a single reference image can provide significant results for contaminated images that are close in time to their acquisition but do not provide effective assistance for other contaminated images. The combination of several reference images can have a positive effect on the results of the reconstruction process.

Concretely, we compare the reconstruction results with 1, 2, and 3 reference images, respectively, and the experimental results relying on the *Bourgogne* dataset are displayed in Table IV. It can be observed that increasing the number of reference images positively influences the performance of the DFTR model, resulting in significant enhancements in both the accuracy and quality of RS image reconstruction.

For the proposed method, using a single reference image can achieve reasonable reconstruction results, but with limited data, the accuracy of the deep domain fidelity is constrained.

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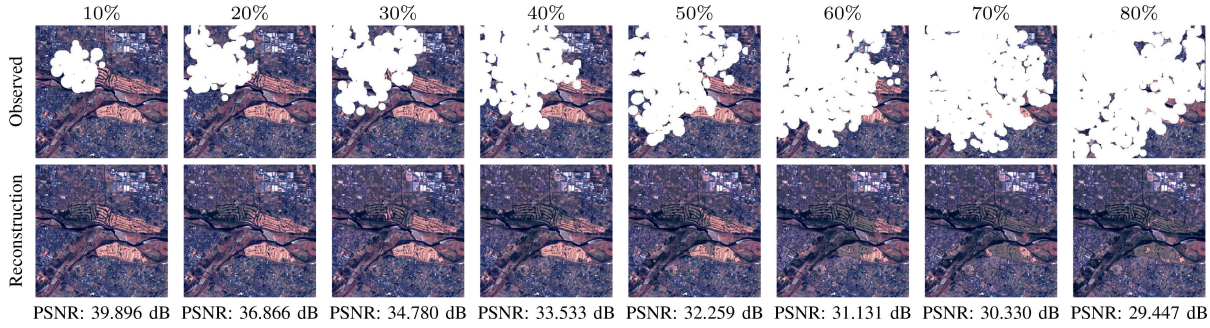


Fig. 16. Observed and corresponding reconstruction results of different cloud coverage ranges (from 10% to 80%).

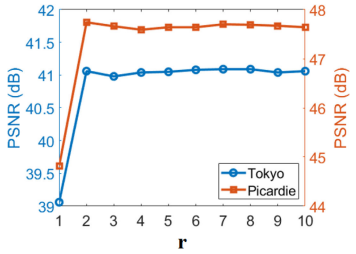


Fig. 17. Influence of the parameters of **r** on different datasets.

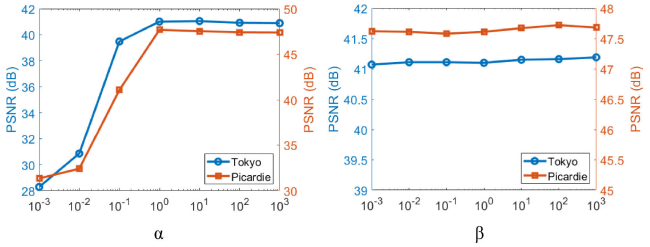


Fig. 18. Sensitivity analysis of the parameters of α and β on different datasets.

Thus, we employ an equal number of reference images and cloud-contaminated images in all of our experiments.

5) Sensitivity Analysis of Parameters: This section examines the impact of three parameters in the DFTR framework, including **r**, α, and β. The parameter **r** is an important parameter in the TR decomposition, reflecting the low-rankness of MTRS images. The weight α balances the significance of the deep domain fidelity term and other terms, and the parameter β determines the weight of the original pixel domain fidelity. Without loss of generality, we perform parametric analyses on the *Tokyo* dataset and the *Picardie* dataset.

We first examine the sensitivity of the parameter **r**. As shown in Fig. 17, the PSNR values corresponding to different values of **r** highlight the consistently superior performance of the proposed DFTR method. The results indicate that the proposed DFTR method consistently demonstrates superior performance. Based on the considerations of the computational complexity and the experimental outcomes, the parameter **r** is set to $(2, 2, 2, 2)$ in the proposed DFTR method in all experiments.

We then analyze the impact of weight α. Fig. 18 illustrates the impact of α on the PSNR value, with α selected from the candidate set $\{0.001, 0.01, 0.1, 1, 10, 100, 1000\}$. The PSNR values increase rapidly as α rises from 0.001 to 1, and when $\alpha = 1$, the PSNR values stabilize. Based on this observation, we empirically set the parameter α to 1 in our method.

In addition, we conduct a series of experiments to discuss the sensitivity of β. Fig. 18 shows the influence of β on the PSNR value, with β chosen from the candidate set $\{0.001, 0.01, 0.1, 1, 10, 100, 1000\}$. The results indicate that the PSNR values remain constant as β increases, demonstrating the robustness of the method to the selection of the parameter β. Based on this observation, we empirically set the parameter β to 1 in the proposed DFTR method.

V. Conclusion

In this article, we have proposed a deep domain fidelity, which can extract invariant deep features to depict the same geographic scene in MTRS images across different time nodes. This method has transcended plain original pixel domain fidelity and addressed the shortcoming that gradient features are generally more fragile and susceptible to distortions. The deep features behind MTRS images are captured through a pretrained deep network, providing better detail preservation for reconstructing cloud-contaminated regions. Leveraging deep domain fidelity, we have suggested a deep feature-guided cloud removal model (called DFTR) for MTRS images, which combines the TR decomposition as the regularization term. In cooperation with LR TR regularization and deep domain fidelity, the proposed DFTR can reconstruct the global spatial–spectral–temporal structure while preserving fine details. The experimental results have shown that the proposed method has outperformed other methods in removing thick clouds, especially in terms of detail preservation. While our proposed unsupervised model shows promising results, there is still potential to improve the DFTR model in terms of running time. In future work, we will attempt to optimize the efficiency of the model to reduce processing time, making it more suitable for deployment in real-world, large-scale scenarios.

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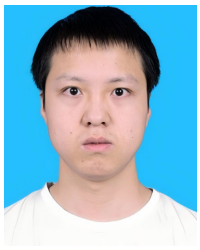
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